Crown And Bridge

Lecture 12

IMPRESSION

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IMPRESSION

- **It is the negative likeness or imprint of the oral soft and hard tissues and their relationship with opposite arches.**

HOW WE CAN GET IMPRESSION

- Impression can be made by placing soft or semi soft material in a tray that is inserted into the patient's mouth. When the material has set, it is removed from the mouth. The "negative" impression is filled with dental stone, and a positive likeness or working cast is obtained.

Requirements of Successful Impression

1- It must be an exact record of all aspects of the prepared tooth, including sufficient unprepared tooth structure adjacent to the finish line.

Requirements of Successful Impression

2- All teeth in the arch and tissues adjacent to the prepared tooth must be reproduced to permit accurate articulation of the cast and allow proper contouring of the

restoration.

Requirements of Successful Impression

3-The impression must be (intact) free of air bubbles, tears, thin spots especially in the area of finish line.

Types of impressions:

1- Primary impression:

it is first impression taken to the patient at the visit of diagnosis using alginate impression material for both arches for the following purposes:

A- Producing study casts to complete case examination and designing of restorations.

B- Construction of special tray if required.

C- Construction of temporary restoration.

Types of impressions:

2- Final impression:

It is taken by elastomeric impression materials (or other materials) after tooth preparation to produce (final) working model on which the final restoration will be constructed.

REQUIREMENTS FOR TAKING FINAL IMPRESSION

For the final impression we need :-

- Special tray.
- Special impression syringe to inject soft impression material .
- Suitable impression material.
- Field isolation and gingival retraction.

The special tray is made on the study cast.

Classification of Impression Materials

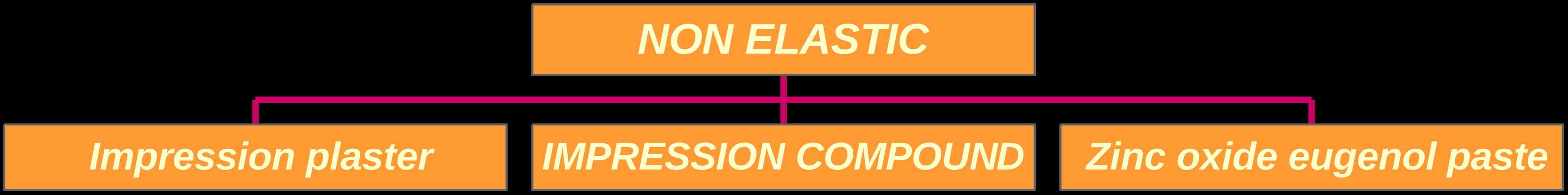
- **IMPRESSION MATERIAL**
  - *NON ELASTIC*
  - *ELASTIC*

Classification of Impression Materials

NON-ELASTIC MATERIALS

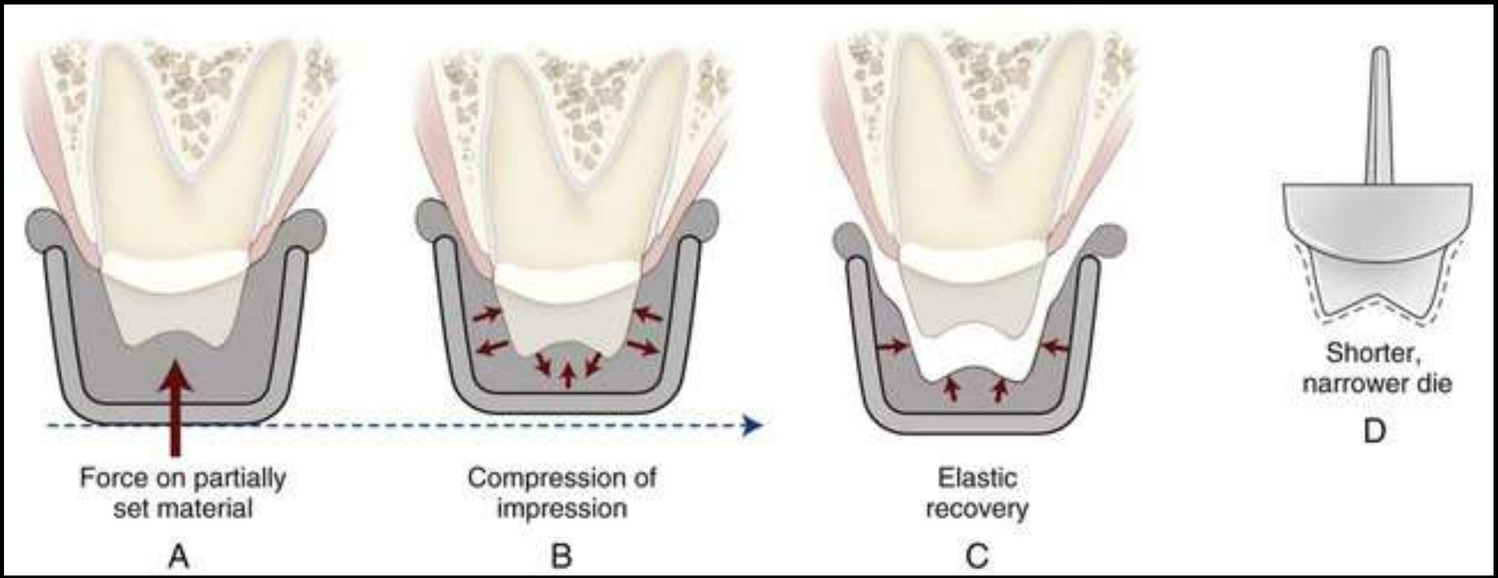
These material used in complete edentulous cases for construction of complete denture as primary or final impressions.

In fixed prosthesis construction its use is very limited for certain cases.



Elastic Impression Material

because it is elastic after setting , so when we remove it after engaging around the undercut area it will not be distorted , and there will be slight temporary deformation (change in shape) that will return to the original shape.



Elastic Impression Materials

Elastic Impression Materials

- Hydrocolloid
- Elastomers Rubber bases

Hydrocolloids

1- Irreversible (Alginate) Alginate is used to produce primary impression ; it doesn’t give us accurate details , so we use it to produce the study cast and articulation casts .

2 - Reversible (Agar- agar) Agar-agar gives accurate details but it needs special tray and extra equipment , so most of the dentists don’t use this type of material although it has accuracy .

Alginate:

powder material mixed with cool water.
It may be regular set or quick set with chromatic setting indicator

Setting reaction of alginate:

When alginate powder is mixed in water a sol is formed which later sets to a gel by a chemical reaction.

First, sodium phosphate reacts with the calcium sulphate to provide adequate working time. After the sodium phosphate is used up, the remaining calcium sulphate reacts with sodium alginate which forms a gel with water.

Agar-agar:

***When agar heated they go into* sol *(liquefy) and on cooling they return back to* gel state.**

Agar-agar :

it requires special and expensive equipments.
Alginate-Agar impression may be used.

Elastomric Impression Material (Rubber Bases)

1- Polysulfide polymer .

2- Silicon.

A- condensation type .

B- Addition type .

3- Polyether .

Different viscosities

- Light body (syringe type)
- Medium body
- Heavy body (tray material)

Monophase (Single-viscosity) Impression Material

- Has a higher apparent standing consistency like heavy body, and yet the same material can have sufficiently high fluidity when injected by a syringe, this material described as **thixotropic**.

Polysulfide

It has better dimensional stability and tear strength than hydrocolloid.

Must be poured as soon as possible after taking the impression

Lead peroxide (catalyst) gives brown color for the material and the material after polymerization is sticky

Properties:

- Un pleasant odor (caused by the high concentration of thiol groups) and un pleasant color. It may stain clothes.
- It has good flexibility (least stiff). Removed from the mouth (and cast) with minimum stress.
- More permanent deformation on removal and less elastic recovery due to reduced degree of polymerization compared to silicone and polyether

Polysulfide properties

It has high tear strength:

(Strain rate or speed of removal) influence the tear resistance and permanent deformation.) snap motion removal for alginate.

Dimensional stability: the stone cast should be poured up immediately (within the first 30 min).

The sources of dimensional changes are:

1- Polymerization shrinkage: during cross-linking most polymers contract slightly.

2- Loss of condensation reaction by-product (water)

3- Absorb fluids if exposed to water, high humidity environment over time.

Factors effect on setting:

1- Heat and humidity accelerates the setting reaction.

2- Base-catalyst ratio. Presence of retarders.

3- Mixing time: More mixing accelerates reaction.

NOTE:

The most accurate impression is made by using custom (special) tray to ensure uniform thickness of the material.

Polysulfide

Wetting

- With hydrophobic material, moisture tends to bead.

- With hydrophilic material, it tends to spreads more uniformly and immediatly.

Polysilicon (Condensation Type)

1- Poor wetting, so it needs dry field during taking an impression and stone pouring.

2- Poor dimensional stability after setting.

Condensation reaction take place with the elimination of ethyl alcohol and water as by-product, this is responsible for the dimensional change (shrinkage) of the material, which result in poor dimensional stability after setting .

Both polysulfide and condensational silicon are condensation polymer.

Polysilicon (Condensation Type)

3- it is more elastic than polysulfide with minimal permanent deformation and recover rapidly when strained.

4- It is not very stiff.

5- Tear resistance is low.

6-Shelf life is limited due to the unstable nature of the Alkyl silicate

Condensation silicone

polysilicon Addition Type

1- Most elastic of the available materials.

2-The lowest permanent deformation and greater dimensional stability. Stiffer than polysulfide.

3-May contain surfactants, which give hydrophilic properties to the material.

4- It is the least affected by pouring delays or by second pours.

5-Mixed without gloves.

Addition silicone

Polyether Impression Material

1- Excellent dimensional stability even when pouring is delayed for prolong periods. The dimensional change is small (no reaction by-product).

2- permanent deformation is low. Recovery is complete because of the excellent elastic properties.

3- The material absorbs moisture that might result in dimensional changes (hydrophilic nature) . It must be kept dry after taking the impression, and some times we can pour it after one day.

- Short setting time.

Polyether Impression Material

4- It is stiff material and we should be very careful when separate it from cast to avoid breaking the area of the prepared tooth.

5- Excellent tear resistance which is better than that of condensation and addition silicone impression material

Impression trays:

Trays are available in different types and sizes with perforations.

1- Ready-made: either plastic or metal (stainless-steel or aluminum).

2- Special tray.

Special tray design

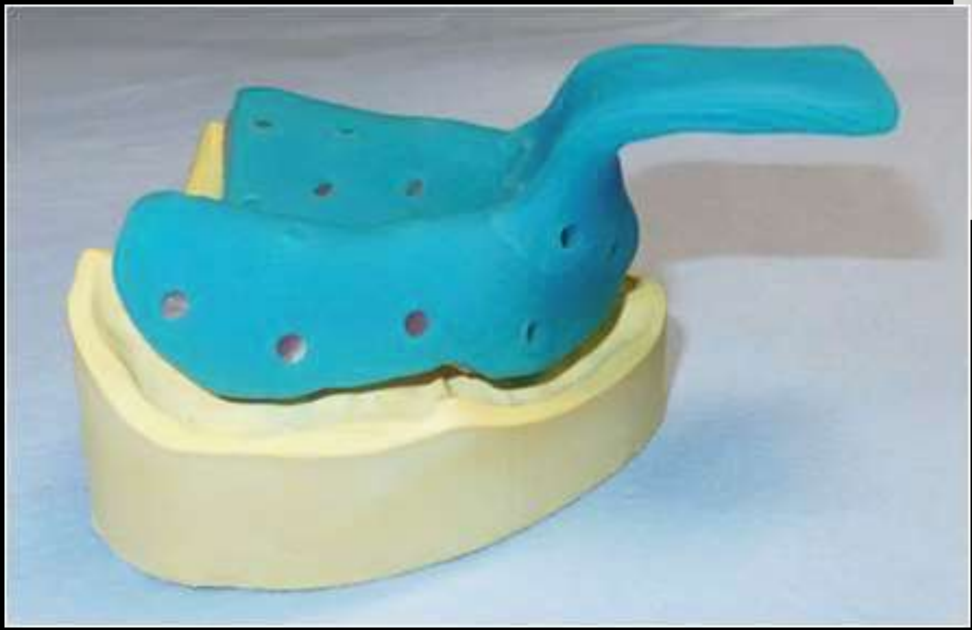
Advantages of Special Tray :-

1- It allows the use of impression material in minimum thickness essential to control the dimensional changes that increased with thick section .

2- It is ready to take impression and no time is consumed for tray selection. It is designed on study cast before the visit of impression.

3- Control the presence of impression material around the working area till the setting of material is completed.

Construction of Special Tray



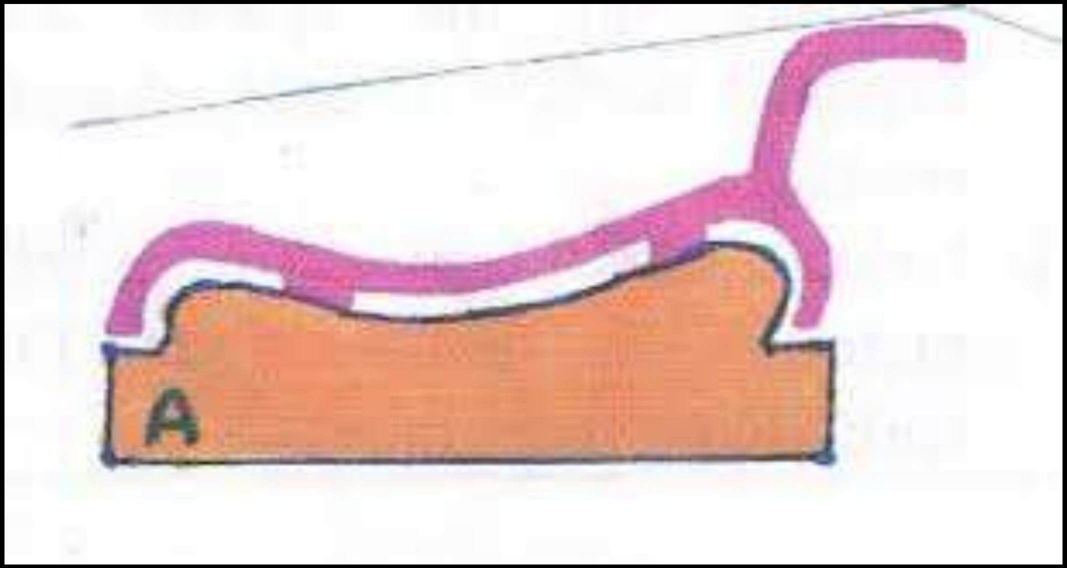
Construction of Special Tray

1- Draw a clear pencil line around the depth of the sulcus to allow the movement of frena.

2- Apply a single wax spacer (layer) accurately on the model and adapt tightly after heat softening.

Construction of Special Tray

3- Open four 1cm square windows within the wax on both sides on teeth which are not prepared to create stoppers in tray latter on.



Construction of Special Tray

4- Mix approximately ½ cup of tray resin for each tray.

5-Apply Vaseline to your hand and adapt the mixed mass on the cast.

Stages which may be noticed during acrylic mixing.

1-Sandy.

2-Stringy.

3- Dough.

4- Rubbery or elastic.

5-Stiff.

Construction of Special Tray

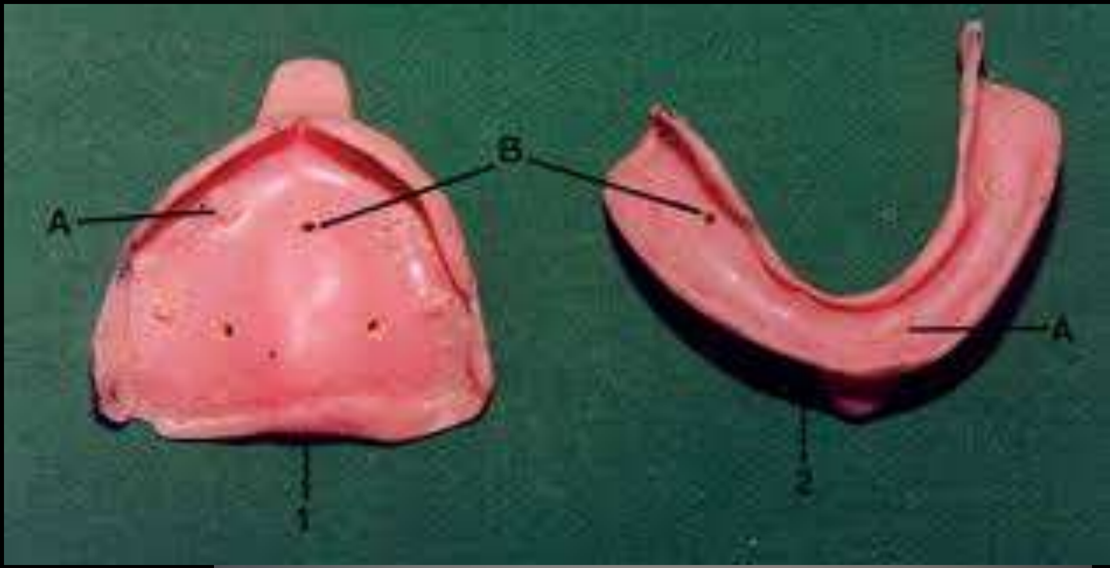
6- Adapting tray material on the waxed cast when the material reached dough stage.

Construction of Special Tray

7- Make a handle for the tray.

8- Peel the softened wax.

9- Do perforations and finish and smooth the acrylic of the tray.



Requirements of Special Tray :-

1- Must be rigid and have thickness of 2-3 mm.

2- Should extend about 4mm cervical below the gingival margin.

3- Stable in the cast with stoppers.

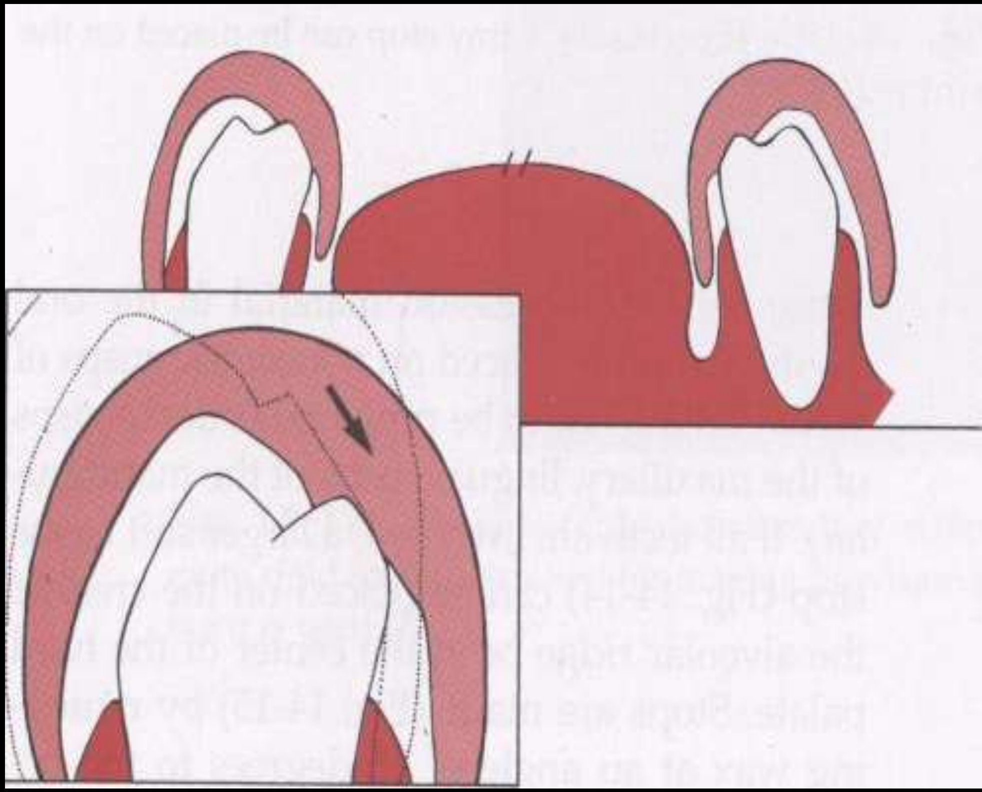
4- Fabricated at least 6 hours prior to its use to allow for complete setting of acrylic material.

Advantages of the Stoppers :-

1- To equalize and stop the pressure that is going to be applied on the tray .

2- It gives us benefits to localize our tray in the mouth when taking impression .

3- Maintain even space for impression material and prevent making contact with the prepared teeth.



IMPRESSION TECHNIQUES

IMPRESSION TECHNIQUES

1. **Single-Step Technique.**
2. **Two-Step Technique.**
3. **Putty Wash Technique.**

Monophase (Single-viscosity) Impression Material

Has a higher apparent standing consistency like heavy body, and yet the same material have sufficiently high fluidity when injected by a syringe, this material also is described as thixotropic.

Single-Step Technique

1- By using material with one viscosity (consistency).

2- Isolate the area and dry the teeth.

3- Load the tray by the material and insert it in the mouth.

4- Remove after setting.

Checking the setting of elastic impression materials

1- Disappearance of stickiness.

2- When pressing or stretching the material it must returned to original shape without distortion or changes.

3- The material left outside the mouth on mixing pad, spatula or nozzle of the injecting pestle must be set according to the points above.

Two-step technique

1- It is done by using two-consistency material with heavy and light bodies. Take the required amount of the heavy body, mix it by hand or using automatic (Auto mix Systems)

Ready-packed cartrages

Automatic dispensers

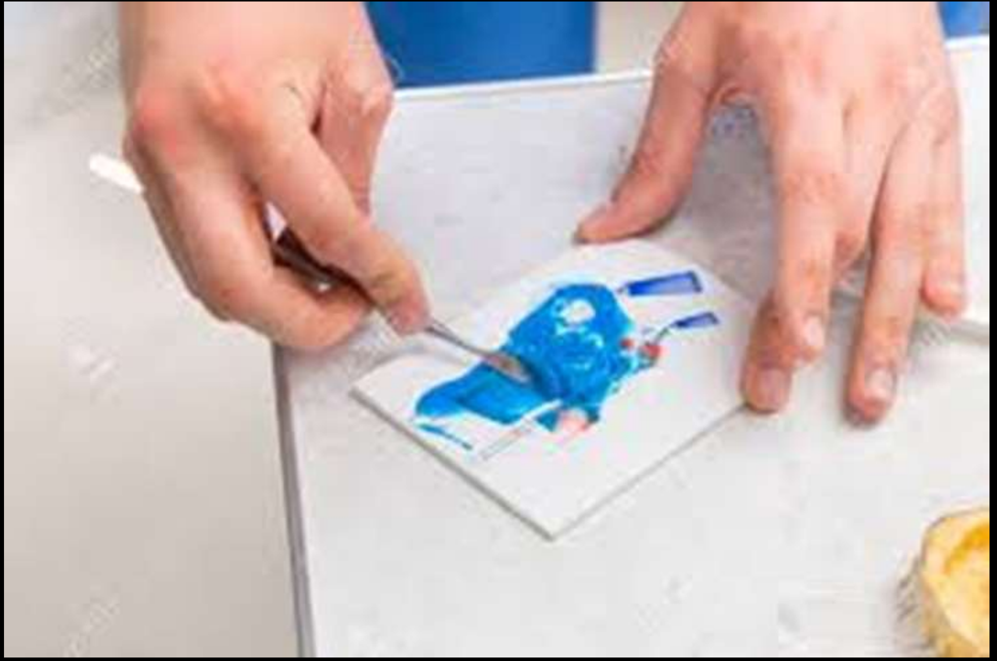
Two-step technique

2- Load mixed heavy body in tray and take impression.

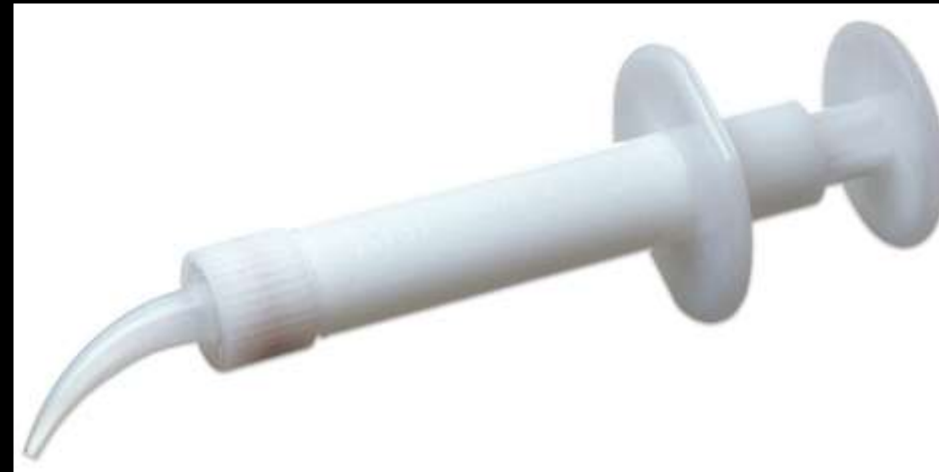
3- Keep tray away and prepare light body.

4- Dispense little amount of light body on mixing pad and mix to homogenous consistency.

5- Pack the impression syringe or use ready-made dispenser syringe.

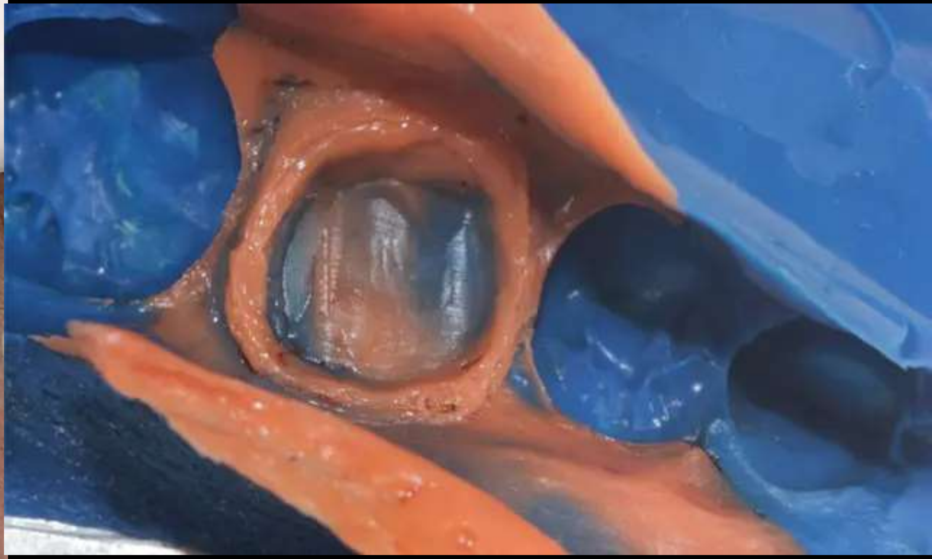


Loading light body



6- Inject light body in the tray on the required area of a bridge or a crown.

7- The left amount injected inside the mouth on abutment teeth after removing traction cord from gingival sulcus and put tray in mouth overall.

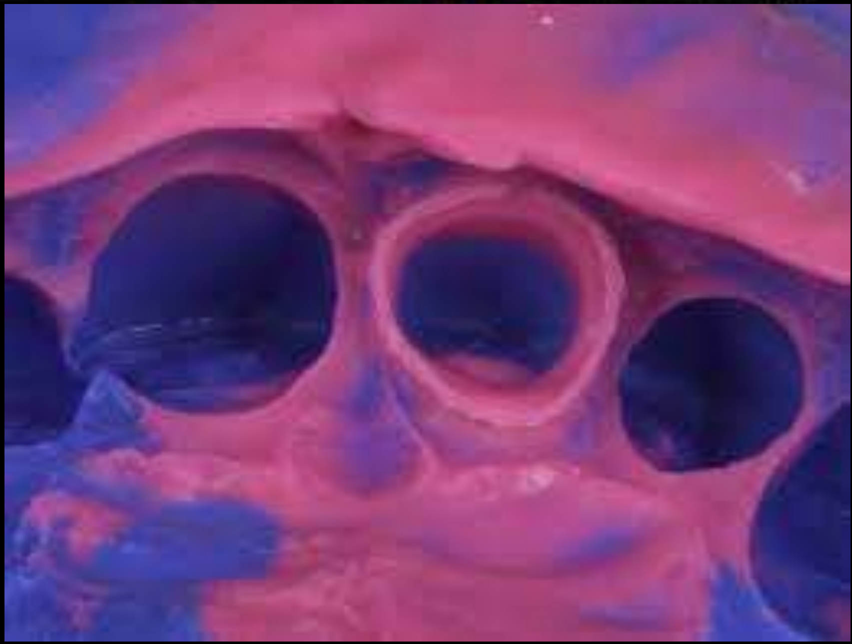


Putty wash techniques:

we use heavy body first before preparation, and the light body left after preparation. The amount of light body used here is very little for reducing dimensional changes.

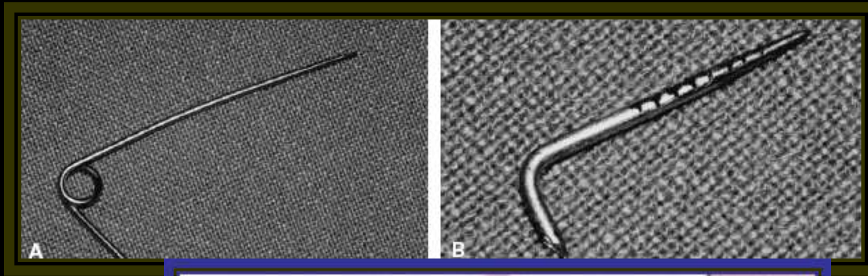
Putty wash techniques

- If we make the impression after preparation.
- We use it with spacer made of polyethylene materials placed over the heavy body.
- And then apply little amount of light body after removing spacer and take impression



Post Crown impression

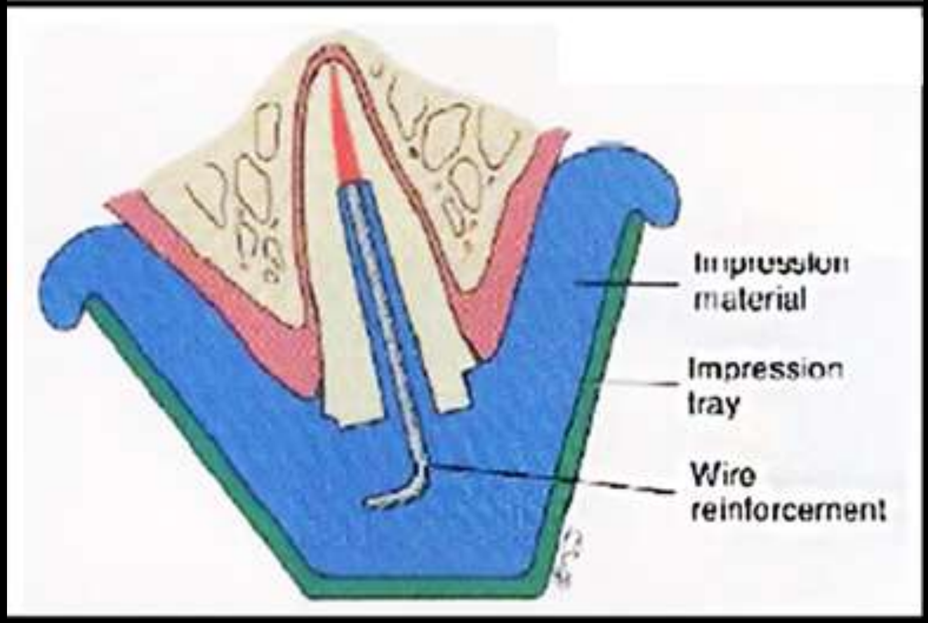
- Tear off or distortion of impression material during removal of impression from teeth is possible.
- So impression material need some type of reinforcement either by a plastic post or by metal wire.
- 1- Prepare L shaped metal wire from paper clip and check its length and placement inside post canal.
- Serrations or roughness is done for retention of impression material to it.



Post crown impression

2- Inject light body inside post space by starting from the end of post and withdraw injection to the coronal opening.

3- Insert the wire inside the post and inject the remaining light body in tray and take overall impression.

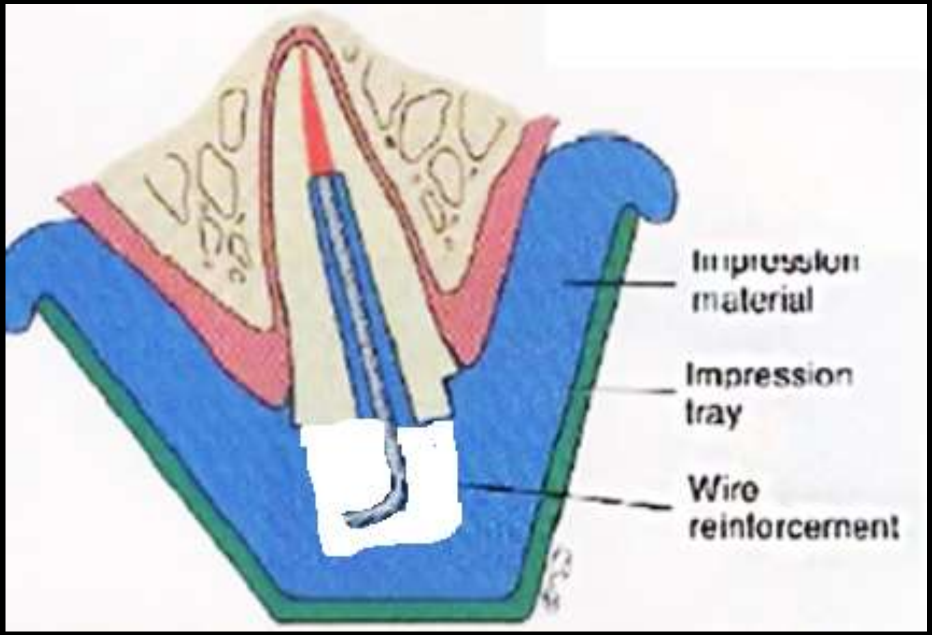


Post crown impression

NOTES:

1- When taking heavy impression remove impression material from the post area allowing space for the wire to allow proper seating of tray.

2- gingival retraction before using light body.



Gingival Retraction(displacement)

It is a procedure used to deflect free marginal gingiva away from the tooth to expose the margin of the preparation (tissue dilation), so that better impression could be taken.

It is used when the margin is placed subgingival or with the level of gingiva.

NEED AND IMPORTANCE OF DISPLACEMENT

1. Adequate access to the prepared tooth.
2. Reproduction of the finish line.
3. For accurate duplicating the sub-gingival margins.
4. Providing the best possible condition for the impression material, fluid control.
5. Precision of the restoration for prevention of periodontal disease.

CLASSIFICATION

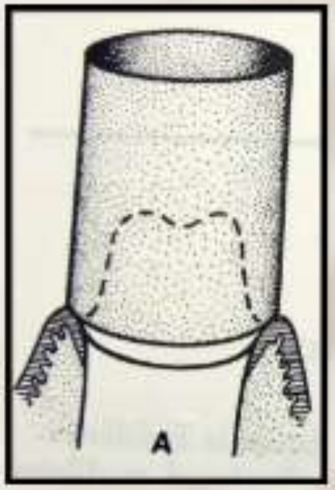
- MECHANICAL
- CHEMICO-MECHANICAL
- ROTARY GINGIVAL CURETTAGE 'GINGETTAGE'
- ELECTROSURGERY
- OTHER METHODS/COMBINATION

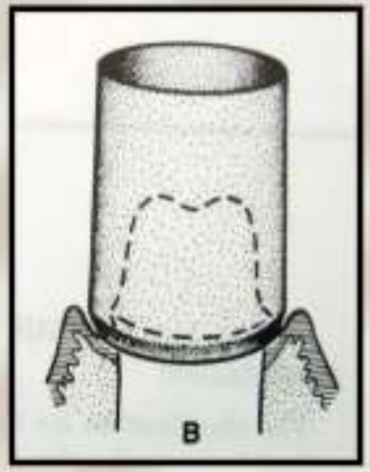
MECHANICAL TISSUE DILATION

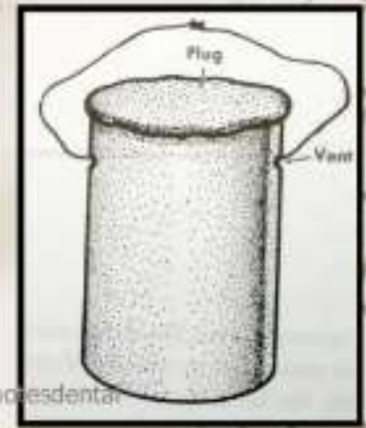
One of the first and earliest methods used for physically displacing the gingiva.

1. Impression material filled copper band/tube
2. Rubber dam
3. Temporary acrylic resin coping
4. Temporary metal crown filled with thermoplastic stopping material
5. Strings or fibers

Impression material filled copper band/tube







VARIOUS IMPRESSION MATERIALS USED:
Impression compound,elastomeric material,
Gutta-percha or auto polymerizing resin.

MECHANICO-CHEMICAL METHODS

- The Mechanical aspect involves placement of a string into the gingival sulcus to displace the tissues.

- The Chemical aspect involves treatment of the string with one or more number of chemical compounds that will induce

i) Temporary shrinkage of the tissues &

ii) Control the hemorrhage & fluid seepage

Mechano-chemical

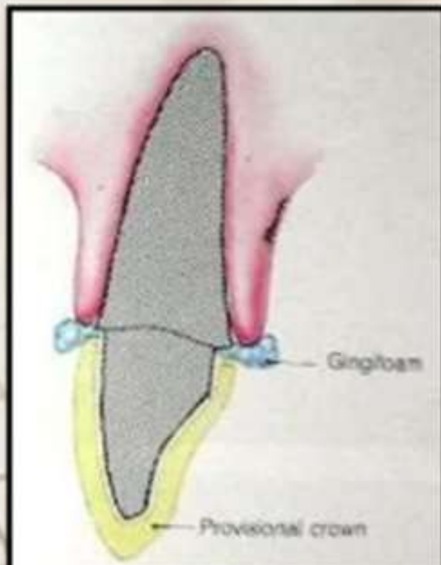
By packing traction cord in the sulcus and leave it for 5-30 minutes when using unsaturated cord. The cord is saturated with vasoconstriction material for shrinking of gingiva and arresting its bleeding which used for 5-10 minutes.

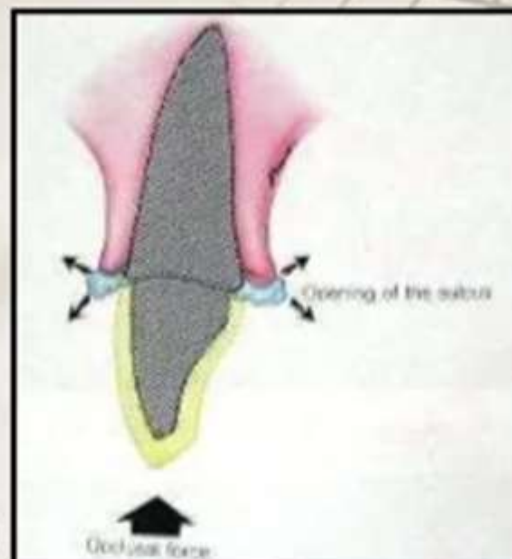
Remove immediately before taking final impression with light body.

ELECTROSURGERY

- Also called 'Troughing' and 'Gingival dilation'
- A trough is created that extends from the crestal height of the gingiva to a point 0.3-0.4mm apical to the finish line using a fully rectified current.

GINGIFOAM


Gingifoam
Provisional crown


Opening of the sulcus
Occlusal force

gingival traction paste

A paste injected around the sulcus for expanding and dilation

1- It is tissue-friendly product.

2- It can expand or increase in size after setting.

3- soluble in water or can be removed easily by air-water spray.

ROTARY GINGIVAL CURETTAGE

- Also called as ‘Gingettage’ and ‘Troughing’
- A technique of using rotary diamond instruments to enlarge the sulcus. It involves preparation of the tooth sub-gingivally while simultaneously curetting the inner lining of the gingival sulcus.
- The goal is to eliminate the trauma from pressure packing and the need for electrosurgical procedures

SUITABILITY OF THE GINGIVA FOR GINGETTAGE

- Absence of bleeding from probing.
- Sulcus depth less than 3 mm.
- Presence of adequate keratinized gingiva.

▪ GingiTrac™

1) Make Matrix

2) Dispense GingiTrac

into the matrix

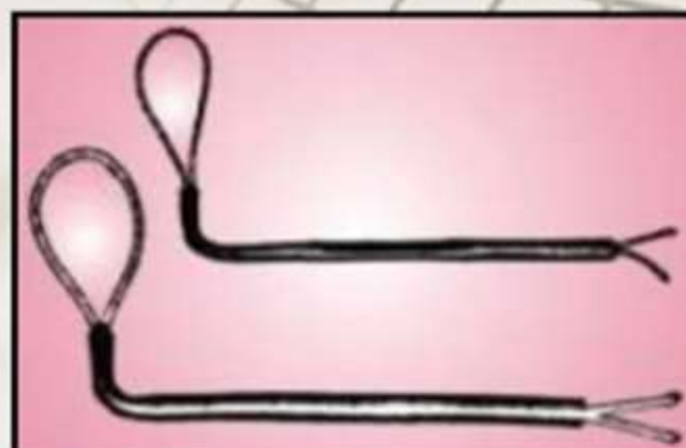
3) Bite down & wait

4) Ready for impression in less than 5 minutes

- **Stay-put**

Stay-put is so pliable that it stays where you put it. Stay-put is a unique combination of softly braided retraction cord and an ultra fine copper filament

- **GINGI-LOOPS**



LASER:

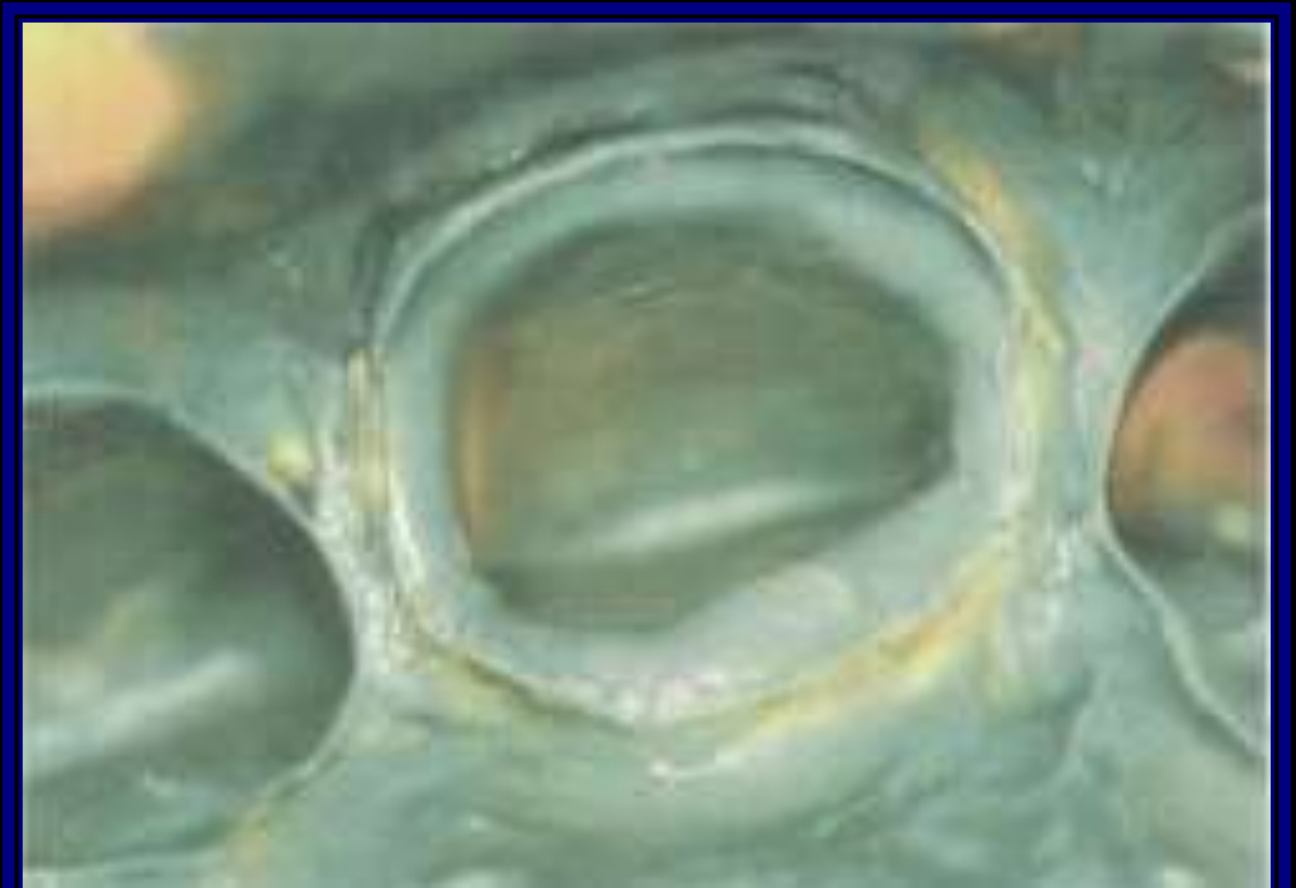
- DIODE AND ND:YAG LASER channels laser through a fiber optic light bundle which incises and cauterizes tissue simultaneously creating haemostasisas well as a retracted field.

PULSED ND = YAG LASER IRRADIATION.

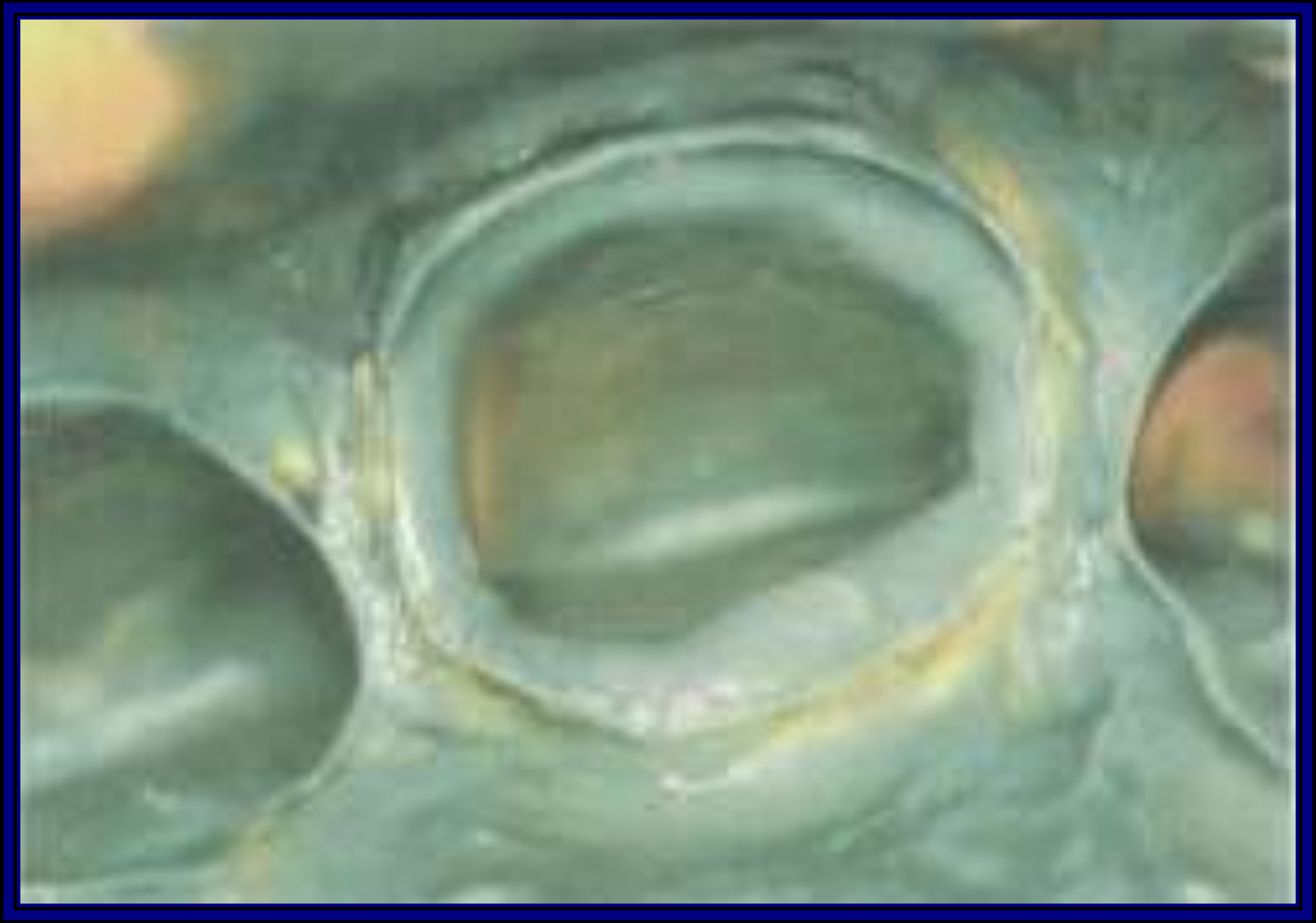
The present histological findings revealed that with the application of PULSED ND: YAG LASER the gingival tissues showed faster healing with less hemorrhage and less inflammatory reaction in comparison with the Ferric sulphate (13.3%).

Inspect the Impression for the Following Points :-

1. Finishing line in the impression must be continuous from one surface to the other.



2. No air bubbles present on the surface of the impression especially at the area of Finishing line.



Inspect the Impression for the Following Points :-

3. The attachment of the impression to the tray must be firm (major difference among impression materials that every one has it is own adhesive that used to bind it to the special tray).

thank

YOU

SO MUCH